

1. Updating Maximum Flow

(a) Capacity increase by 1

Edge (u, v) capacity: $c(u, v) \rightarrow c(u, v) + 1$. Flow f still feasible (capacity only increased).

In the residual graph G_f with updated capacity, the residual capacity of (u, v) increases by 1. Find one augmenting s - t path via BFS.

- If found: augment by 1 $\implies$ new max flow = $|f| + 1$.
- If not found: $|f|$ unchanged (still max).

$$\boxed{O(m + n): \text{ one BFS}}$$

(b) Capacity decrease by 1

Edge (u, v) capacity: $c(u, v) \rightarrow c(u, v) - 1$.

Case 1: $f(u, v) \leq c(u, v) - 1$. Flow still feasible under new capacity. Still max (no new augmenting paths). Done.

Case 2: $f(u, v) = c(u, v)$ (saturated). Set $f(u, v) \leftarrow c(u, v) - 1$. Now u has excess $+1$, v has deficit -1 .

Build residual graph with updated capacity and flow. Find augmenting path from u to v :

- If found: push 1 unit $\implies$ conservation restored, $|f|$ unchanged.
- If not found: max flow = $|f| - 1$. The original flow decomposition includes an s - t path through (u, v) . Remove 1 unit of flow along this entire path: BFS from u backward along flow-carrying edges to s (reduces inflow to u , fixing excess), BFS from v forward along flow-carrying edges to t (reduces outflow from v , fixing deficit). Result: valid flow of value $|f| - 1$.

$$\boxed{O(m + n): \text{ constant number of BFS calls}}$$

2. Minimum Cut in Undirected Graphs

(a)

Replace each undirected edge $\{a, b\}$ with two directed edges (a, b) and (b, a) , each capacity 1. For every pair $s, t \in V$: compute max s - t flow = min s - t cut.

$$\text{Global min cut} = \min_{s, t \in V} \text{min } s\text{-}t \text{ cut}$$

$$\binom{n}{2} \text{ max-flow computations } \implies \boxed{O(n^2 \cdot T_{\text{maxflow}})}$$

(b)

Fix an arbitrary vertex $s \in V$. For each $t \in V \setminus \{s\}$: compute max s - t flow.

$$\text{Global min cut} = \min_{t \neq s} \min s\text{-}t \text{ cut}$$

Correctness: Let (A, B) be a global min cut. $s \in A$ or $s \in B$; WLOG $s \in A$. $\forall t \in B$:

$$\min s\text{-}t \text{ cut} \leq |(A, B)| = \text{global min cut} \quad ((A, B) \text{ is an } s\text{-}t \text{ cut})$$

$$\min s\text{-}t \text{ cut} \geq \text{global min cut} \quad (\text{every } s\text{-}t \text{ cut is a cut})$$

$\implies$ equality. Since we try all t , some $t \in B$ is tested.

$n - 1$ max-flow computations. With Ford-Fulkerson ($O(mF)$ per flow, $F = \text{max flow value}$):

$$\boxed{O(n \cdot mF)}$$

3. Maximal versus Maximum Matchings

(a)

Let M_1 be maximal, M_2 be maximum in bipartite $G = (X \cup Y, E)$.

$|M_1| \leq |M_2|$: M_2 is maximum $\implies |M_2| \geq |M|$ for every matching M .

$|M_2| \leq 2|M_1|$: M_1 maximal $\implies \forall e \in E$: e has ≥ 1 endpoint in $V(M_1)$. In particular, $\forall e \in M_2$: e has ≥ 1 endpoint in $V(M_1)$. Since $|V(M_1)| = 2|M_1|$ and M_2 is a matching (each vertex in ≤ 1 edge):

$$|M_2| \leq |V(M_1)| = 2|M_1| \quad \square$$

$$\boxed{|M_1| \leq |M_2| \leq 2|M_1|}$$

(b)

For $0 < k \leq \ell \leq 2k$, let $d = \ell - k$ (so $0 \leq d \leq k$).

Construction:

- $k - d$ isolated edges (x_i, y_i) for $i = 1, \dots, k - d$. In both matchings.
- d augmenting paths of length 3: for $j = 1, \dots, d$, path $x'_{2j-1} - y'_{2j-1} - x'_{2j} - y'_{2j}$.

Maximal matching M_1 (size k): Take isolated edges $(k - d)$ + middle edge of each P_3 (d).
Maximal: in each P_3 , both outer edges have a saturated endpoint.

Maximum matching M_2 (size ℓ): Take isolated edges $(k - d)$ + two outer edges of each P_3 ($2d$).
Total = $k - d + 2d = k + d = \ell$.

$$\boxed{|M_1| = k, \quad |M_2| = \ell}$$

(c)

$X = \{x_1, x_2\}$, $Y = \{y_1, \dots, y_8\}$. Edges: (x_1, y_j) and (x_2, y_j) for $j = 1, \dots, 5$.

$|V| = 10$, $|E| = 10$. Maximum matching = 2 (since $|X| = 2$).

Every maximal matching has size 2: if $|M| = 1$, say $M = \{(x_1, y_j)\}$, then x_2 is free and $\exists y_i$ ($i \neq j$, $i \leq 5$) free with $(x_2, y_i) \in E \implies M$ not maximal. Symmetric for $M = \{(x_2, y_j)\}$.

Every maximal matching is maximum (size 2)

4. Symmetric Difference of Two Matchings

Let M_1, M_2 be matchings in $G = (V, E)$.

(a)

Consider the graph $H = (V, M_1 \oplus M_2)$. Each vertex v is in ≤ 1 edge of M_1 and ≤ 1 edge of M_2 (matching property) $\implies \deg_H(v) \leq 2$.

A graph with $\Delta \leq 2$ is a disjoint union of paths and cycles. $\square$

(b)

In any cycle of $M_1 \oplus M_2$, consecutive edges share a vertex. Since each matching uses each vertex at most once, consecutive edges must come from different matchings ($M_1 \setminus M_2$ and $M_2 \setminus M_1$ alternate). An alternating cycle has even length. $\square$

(c)

By (a) and (b), $M_1 \oplus M_2$ decomposes into paths and even cycles. In each component, count M_1 -edges minus M_2 -edges:

- Even cycle: alternating $\implies$ equal count $\implies$ contribution = 0.
- Even path: alternating $\implies$ equal count $\implies$ contribution = 0.
- Odd path: one matching has 1 more edge $\implies$ contribution = ± 1 .

Total: $|M_1 \setminus M_2| - |M_2 \setminus M_1| = |M_1| - |M_2| \neq 0$. Only odd paths contribute nonzero terms $\implies \exists$ at least one odd path. $\square$